We have

$$h_i = t \sum_{j \neq i}^{N} \delta_{ij}$$

where $H = -\sum_i < \sum_j J_{ij} (\sigma_i \sigma_j + t(\sigma_i + \sigma_j))$

t is now generator for magnetisation

Normally magnetisation $= < \frac{1}{N} \sum_i \sigma_i >$

but we want to know specific aligned spins

$$f = \frac{1}{N} \ln \underbrace{Tr_N \left(e^{-\beta H} \right)}_{Z} = \text{free energy}$$

Measure magnetisation as $\frac{\partial f}{\partial t}$

$$= < \sum_i \sigma_i \sum_j J_{ij} > =$$

$$\frac{1}{N} \frac{Tr_N \, \partial/\partial t \, e^{-\beta H}}{Tr_N \, e^{-\beta H}} = \frac{1}{N} \sum_i \sigma_i \sum_j J_{ij}$$

$$[Z^n] = \mathrm{Tr}_{n,N} \int \left(\prod_{i<j} \prod dJ_{ij} P(J_{ij}) \right)$$

$$\left[\prod_{i<j} \prod \exp\left\{ \beta \sum_{\alpha} \underbrace{\left(\sigma_i^\alpha \sigma_j^\alpha + t(\sigma_i^\alpha + \sigma_j^\alpha) \right)}_{q_{ij}} \right\} \right]$$

$$= \mathrm{Tr}_{n,N} \prod_{i<j} \prod \int dx\, P(x)\, e^{x\beta q_{ij}}$$

$$P(x) = \sqrt{\frac{N}{2\pi}} \underbrace{\exp\left(\frac{-N}{2} \left(x - \frac{J_0}{N} \right)^2 \right)}$$

$$\int dx\, P(x) e^{x\beta q_{ij}}:$$

$$\sqrt{\frac{N}{2\pi}} \int x^2 - 2\frac{J_0}{N} + x\beta q$$

$$-\frac{N}{2} x^2 + J_0 x + \frac{-J_0^2}{2N} + x\beta q$$

$$= -\frac{N}{2} \left(x - \frac{J_0 \beta q}{N} \right)^2 + \frac{1}{2N} \left(J_0 + \beta q \right)^2 - \frac{J_0^2}{2N}$$

$$\underbrace{\qquad\qquad}$$

$$1 \exp\left\{ \frac{1}{N} J_0 \beta q + \frac{\beta^2}{2N} q^2 \right\}$$

Now we can substitute back in.

$$[Z^n] = \mathrm{Tr}_{\sigma,\alpha} \exp\left\{ \sum_{i<j} \left[\frac{1}{N} \beta J_0 \delta(r_i - r_j) \right. \right.$$

$$\left. \left. + \frac{\beta^2}{2N} \right. \right.$$

$$\sum_{i<j} = \frac{1}{2} \sum_i \sum_j - \frac{1}{2} \sum_{i=j}$$

So we end up with $\sum_{i<j} g_{ij}^2 =$

$$= \frac{1}{2} \sum_\alpha \sum_\gamma \left\{ \sum_i \sum_j \left(\sigma_i^\alpha \sigma_j^\alpha \sigma_i^\gamma \sigma_j^\gamma \right) t \right.$$

$$\sigma_i^\alpha \sigma_j^\gamma) \sigma_i^\gamma + t (\sigma_a^i \sigma_\lambda^i) \sigma_i^\gamma \sigma_j^\gamma$$

$$+ t^2 (\sigma_i^\alpha + \sigma_j^\alpha)(\sigma_i^\gamma + \sigma_j^\gamma)$$

$$= \frac{1}{2} \sum_\alpha \sum_\gamma \left\{ N^2 Q_{\alpha\gamma}^2 + 4t N^2 Q_{\alpha\gamma} M_\alpha \right.$$

$$+ 2t^2 N^2 (Q_{\alpha\gamma} + \text{Memes})$$

$$\left. - N - 4tNM_\alpha - 4t^2 N Q_{\alpha\gamma} \right)$$

Compressed infinitely many sums into one sum

$t=0$ reduces to original Sherrington Kirkpatrick model

Compute square on 1st 2 terms:

$$Q^2 + 4tQM = Q^2 + 2Q(2M) + (2tM)^2 - (2tM)^2$$

$$2Q(2tM) = (Q+2tM)^2 - 4t^2M^2$$

When summing over α, γ:

$$\sum_\alpha \sum_\gamma \left((Q_{\alpha\gamma} + 2tM_\alpha)^2 - \underbrace{4t^2 n \sum_\alpha M_\alpha^2}_{O(n^2)} \right)$$

can ignore

Now we go back to

$$\sum_i c \sum_j q_{ij} = \frac{1}{2} \sum_i \sum_j q_{ij} - \frac{1}{2} \sum_i q_{ii}$$

$$= \frac{1}{2} \sum_\alpha \sum_i \sum_j \left(\sigma_i^\alpha \sigma_j^\alpha + t(\sigma_i^\alpha + \sigma_j^\alpha) \right) - \frac{1}{2} \sum_i (\sigma_i^{\alpha 2} + 2t\sigma_i^\alpha)$$

$$= \frac{1}{2} \sum_\alpha \left[N^2 M_\alpha^2 + 2N^2 t M_\alpha - \frac{N}{2} - 2tNM_\alpha \right]$$

$$= \frac{N^2}{2}\left\{ \sum_\alpha M_\alpha^2 + 2t \sum_\alpha M_\alpha - \frac{n}{N} - t\cancel{N^2 \sum_\alpha M_\alpha} \right.$$

$\cancel{N^2}$ → small

$$= \frac{N^2}{2}\left(\sum_\alpha M_\alpha^2 + 2t \sum_\alpha M_\alpha - \frac{n}{N} \right)$$

We can now write

$$[Z^n] = Tr_{n,N} \exp\left\{ \frac{\beta^2}{4N} \sum_\alpha \sum_\gamma \left[N^2 (Q_{\alpha\gamma} + 2tM_\alpha)^2 \right.\right.$$

$$+ 2t^2 N^2 (Q_{\alpha\gamma} + M_\alpha M_\gamma) - N$$

$$- 4tNM_\alpha - 4t^2 N Q_{\alpha\gamma}$$

$$\left.\left. + \frac{N^2}{2} \frac{J_0 \beta}{N} \left[\sum_\alpha M_\alpha^2 + 2t \sum_\alpha M_\alpha - \frac{N}{n} \right] \right\}$$

where $Q_{\alpha\gamma} = \frac{1}{N} \sum_i \sigma_i^\alpha \sigma_i^\gamma$, $M_\alpha = \frac{1}{N} \sum_i \sigma_i^\alpha$

for $\alpha = \gamma$

$$= Tr_{n,N} \exp\left\{ \frac{\beta^2}{2N} \sum_\alpha \sum_{r \neq \alpha} \{ \cdots \} + \frac{\beta^2}{4N} \sum_\alpha \left(\alpha^2 \underbrace{(Q_{\alpha\alpha}}_{\to 1} \right.\right.$$

$$\left. + 2tM_\alpha)^2 + 2t^2 N^2 (1 + M_\alpha^2) - \cancel{N} + \right.$$

$$\left.\left. \cancel{4tNM_\alpha} - \cancel{4t^2 N} \right\} \right.$$

$$= \mathrm{Tr}_{n,N} \exp\left\{ \frac{\beta^2 N}{2} \sum_{\alpha<\gamma} \sum \left\{ (Q_{\alpha\gamma} + 2tM_\alpha)^2 \right.\right.$$

$$\left. + 2t^2 (Q_{\alpha\gamma} + M_\alpha M_\alpha) - \frac{1}{N} + \frac{4tM_\gamma}{N} - \frac{4t^2 Q_{\alpha\gamma}}{N} \right\}$$

missing diagonal terms

$$\left. + \frac{N\beta J_0}{2} \left\{ \sum_\alpha M_\alpha^2 + 2t \sum_\alpha M_\alpha - \frac{n}{N} \right\} \right\}$$

$$= \mathrm{Tr}_{n,N} \exp\left[\frac{\beta^2 N}{2} \sum_{\alpha<\gamma} \sum \left\{ \left(Q_{\alpha\gamma} + t(M_\alpha + M_\gamma) \right)^2 \right.\right.$$

$$\left. + 2t^2 Q - \frac{1}{N} - \frac{4t}{N} M_\gamma \right\}$$

correct $+ \frac{N\beta^2}{4} \sum_\alpha \left(4t^2 M_\alpha^2 + 4tM_\alpha + (2t^2+1) \right)$

$$\left. + \frac{N\beta J_0}{2} \left\{ \sum_\alpha M_\alpha^2 + 2t \sum_\alpha M_\alpha - \frac{n}{N} \right\} \right\}$$